INTERNATIONAL SEARCH REPORT

International application No
PCT/US2022/011606

A. CLASSIFICATION OF SUBJECT MATTER
INV. G01N33/50 G01N33/574 G01N33/58
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G01N

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X,P	BUDKA JUSTIN ET AL: "Abstract CT166: Pretreatment (PreTx) immune cell phenotypes in peripheral blood associated with the tumor immune contexture, product attributes, and durable clinical efficacy in patients with large B-cell lymphoma (LBCL) treated with axicabtagene ciloleucel (axi-cel)", CANCER RESEARCH, vol. 81, no. 13 SUPPl, 1 July 2021 (2021-07-01), page CT166, XP055915524, US ISSN: 0008-5472, DOI: 10.1158/1538-7445.AM2021-CT166 Retrieved from the Internet: URL:https://aacrjournals.org/cancerres/article/81/13_Supplement/CT166/669782/Abstract-CT166-Pretreatment-PreTx-immune-cell> -/--	1-27

[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
29 April 2022	12/05/2022

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Gonçalves Mauger, M

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Form PCT/ISA/210 (second sheet) (April 2005)

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
	abstract -----	
X,P	WO 2021/168375 A1 (KITE PHARMA INC [US]) 26 August 2021 (2021-08-26) paragraph [0020] - paragraph [0030] -----	1-27
X	WO 2019/232510 A1 (KITE PHARMA INC [US]) 5 December 2019 (2019-12-05) claims -----	1-27
X	PETERSEN CHRISTOPHER T ET AL: "Improving T-cell expansion and function for adoptive T-cell therapy using ex vivo treatment with PI3K[delta] inhibitors and VIP antagonists", BLOOD ADVANCES, vol. 2, no. 3, 13 February 2018 (2018-02-13), pages 210-223, XP055915745, ISSN: 2473-9529, DOI: 10.1182/bloodadvances.2017011254 page 210 - page 211 page 217 - page 219; figures -----	1-27
X	LOCKE FREDERICK L. ET AL: "Tumor burden, inflammation, and product attributes determine outcomes of axicabtagene ciloleucel in large B-cell lymphoma", BLOOD ADVANCES, vol. 4, no. 19, 13 October 2020 (2020-10-13), pages 4898-4911, XP055849335, ISSN: 2473-9529, DOI: 10.1182/bloodadvances.2020002394 Retrieved from the Internet: URL:https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7556133/pdf/advancesADV2020002394.pdf> page 4899 - page 4909; figures -----	1-27
X	JOSEPH A. FRAIETTA ET AL: "Determinants of response and resistance to CD19 chimeric antigen receptor (CAR) T cell therapy of chronic lymphocytic leukemia", NATURE MEDICINE, vol. 24, no. 5, 30 April 2018 (2018-04-30), pages 563-571, XP055542305, New York ISSN: 1078-8956, DOI: 10.1038/s41591-018-0010-1 page 563 -----	1-27
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C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2019/213282 A1 (NOVARTIS AG [CH]; UNIV PENNSYLVANIA [US]) 7 November 2019 (2019-11-07) claims -----	1-27
X	MICHAEL D. JAIN ET AL: "Concise Review: Emerging Principles from the Clinical Application of Chimeric Antigen Receptor T Cell Therapies for B Cell Malignancies : Principles of CAR T Therapy", STEM CELLS, vol. 36, no. 1, 25 October 2017 (2017-10-25), pages 36-44, XP055607326, ISSN: 1066-5099, DOI: 10.1002/stem.2715 see introduction; page 36 - page 37 -----	1-27

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Box No. I Nucleotide and/or amino acid sequence(s) (Continuation of item 1.c of the first sheet)

1. With regard to any nucleotide and/or amino acid sequence disclosed in the international application, the international search was carried out on the basis of a sequence listing:

   a. [ ] forming part of the international application as filed:

      [ ] in the form of an Annex C/ST.25 text file.

      [ ] on paper or in the form of an image file.

   b. [ ] furnished together with the international application under PCT Rule 13ter.1(a) for the purposes of international search only in the form of an Annex C/ST.25 text file.

   c. [X] furnished subsequent to the international filing date for the purposes of international search only:

      [X] in the form of an Annex C/ST.25 text file (Rule 13ter.1(a)).

      [ ] on paper or in the form of an image file (Rule 13ter.1(b) and Administrative Instructions, Section 713).

2. [ ] In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that forming part of the application as filed or does not go beyond the application as filed, as appropriate, were furnished.

3. Additional comments:

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/US2022/011606

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 2021168375 A1	26-08-2021	US 2022016165 A1	20-01-2022
		WO 2021168375 A1	26-08-2021
WO 2019232510 A1	05-12-2019	AU 2019276592 A1	10-12-2020
		CA 3101856 A1	05-12-2019
		CN 112512537 A	16-03-2021
		EP 3801572 A1	14-04-2021
		JP 2021525863 A	27-09-2021
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		WO 2019232510 A1	05-12-2019
WO 2019213282 A1	07-11-2019	EP 3788369 A1	10-03-2021
		US 2021396739 A1	23-12-2021
		WO 2019213282 A1	07-11-2019